q-DE RHAM COHOMOLOGY AS A MOTIVIC SPECTRUM

NIKLAS ARPPE

Abstract. We construct a motivic spectrum q-dR^{mot} in Annala–Hoyois–Iwasa's category of non-$\mathbf{A}^1$-invariant motivic spectra representing global q-de Rham cohomology. Moreover, we show that this is compatible with the motivic spectra representing de Rham and prismatic cohomology. The main novelty is that we make no reference to orientations, but rather construct the motivic spectrum by hand.

1. Introduction

Global q-de Rham cohomology is a functorial $(q-1)$-adic deformation of de Rham cohomology of smooth $\mathbf{Z}$-schemes. It was conjectured to exist by Scholze [Sch16] and constructed via prismatic cohomology by Bhatt–Scholze and Wagner [BS22; Wag25]. For a smooth $\mathbf{Z}$-scheme X it takes values in derived $(q-1)$-complete $\mathbf{Z}[\![q-1]\!]$-modules, reduces to $R\Gamma(X, \Omega^{\bullet}_{X/\mathbf{Z}})$ modulo $(q-1)$, and its p-completion is prismatic cohomology relative to the q-de Rham prism $(\mathbf{Z}_p[\![q-1]\!], ([p]_q))$. Like de Rham cohomology over $\mathbf{Z}$, it is not $\mathbf{A}^1$-invariant, so it cannot be represented in Morel–Voevodsky's stable motivic homotopy category.

Annala, Hoyois and Iwasa have developed a theory of motivic spectra without $\mathbf{A}^1$-invariance [AI23; AHI24a; AHI24b], in which $\mathbf{A}^1$-invariance is replaced by smooth blow-up excision and $\mathbf{P}^1$ is inverted directly. Many p-adic cohomology theories are representable there, including syntomic, prismatic and crystalline cohomology [AHI24b, Section 6], and de Rham cohomology together with its Hodge filtration [AP26]. These constructions use an orientation: a theory of first Chern classes, which yields a projective bundle formula and thereby both blow-up excision and the $\mathbf{P}^1$-periodicity. The goal of this paper is to show that global q-de Rham cohomology is representable as well, without using any orientation.

Theorem A (Theorem 4.7)**.** — *There is a motivic spectrum q-dR$^{\mathrm{mot}} \in \mathrm{MS}_{\mathbf{Z}}$ representing global q-de Rham cohomology. More precisely, let $\mathbf{L} := \widetilde{H}^2_{q\text{-dR}}(\mathbf{P}^1)$, which is a free $\mathbf{Z}[\![q-1]\!]$-module of rank one. Then for every smooth $\mathbf{Z}$-scheme X and all $n, m \in \mathbf{Z}$,*

$$\pi_{2m-n} \operatorname{map}_{\mathrm{MS}_{\mathbf{Z}}}(X_+, \Sigma^m_{\mathbf{T}}\, q\text{-dR}^{\mathrm{mot}}) \cong H^n_{q\text{-dR}}(X) \otimes_{\mathbf{Z}[\![q-1]\!]} \mathbf{L}^{\otimes -m}.$$

We prove this by a direct construction. A recurring theme is the application of the derived Nakayama lemma to reduce mod $(q-1)$, which allows us to perform hands-on arguments involving the simpler de Rham cohomology.

Theorem B (Proposition 5.3, Proposition 5.7, Theorem 5.8)**.** —

(1) The cofiber of $q-1$ on q-dR^{mot} represents de Rham cohomology, and agrees with the de Rham spectrum of Annala–Pstrągowski.

(2) For every prime p, the p-completion $(q\text{-dR}^{\mathrm{mot}})^{\wedge}_p$ represents prismatic cohomology relative to the q-de Rham prism (A, I), and agrees with the motivic spectrum $\mathrm{H}A^{\triangle}$ of Annala–Hoyois–Iwasa. In particular, $\mathbf{L}^{\wedge}_p \cong A\{-1\}$ is the Breuil–Kisin twist.

(3) There is a cartesian square in $\mathrm{MS}_{\mathbf{Z}}$

$$\begin{array}{ccc} q\text{-}\mathrm{dR}^{\mathrm{mot}} & \longrightarrow & \prod_p (q\text{-}\mathrm{dR}^{\mathrm{mot}})^\wedge_p \\ \downarrow & & \downarrow \\ q\text{-}\mathrm{dR}^{\mathrm{mot}}_{\mathbf{Q}} & \longrightarrow & q\text{-}\mathrm{dR}^{\mathrm{mot}}_{\mathbf{A}}, \end{array}$$

exhibiting q-$\mathrm{dR}^{\mathrm{mot}}$ *as glued from the prismatic motivic spectra and a rational corner representing* $R\Gamma_{\mathrm{dR}}(X_{\mathbf{Q}}/\mathbf{Q})[\![q-1]\!]$.

In this sense, q-$\mathrm{dR}^{\mathrm{mot}}$ is a q-deformation of the de Rham motivic spectrum, and it glues the prismatic motivic spectra for all primes p, constructed without reference to Chern classes. An orientation of q-$\mathrm{dR}^{\mathrm{mot}}$ would trivialize $\mathbf{L}$. Wagner has suggested that the q-logarithm should provide q-de Rham Chern classes; we hope to return to this.

1.1. **Outline.** In Section 2 we recall global q-de Rham cohomology and establish the properties we need: descent, multiplicativity, the Künneth formula, and the computation for $\mathbf{P}^1$. In Section 3 we recall the category MS_S and explain how to recognize motivic spectra from sequential data. Section 4 contains the construction of q-$\mathrm{dR}^{\mathrm{mot}}$ and the proof of the first theorem. Finally, in Section 5 we compare q-$\mathrm{dR}^{\mathrm{mot}}$ with de Rham and prismatic cohomology and prove the fracture square.

1.2. **Notation and conventions.** By a category we will always mean an $(\infty,1)$-category, and we follow the conventions of Lurie regarding them [Lur09; Lur17; Lur18]. If $I \subseteq A$ is an ideal in a ring, then we denote by $D(A)^\wedge_I \subseteq D(A)$ the full subcategory of derived I-complete objects in the derived category of A. By a scheme we mean a quasi-compact and quasi-separated scheme. All of our rings and schemes will be classical, i.e., neither derived nor animated.

2. q-de Rham cohomology

In this section, we recall some essential properties of global q-de Rham cohomology. For a more complete account (including the p-adic story), we refer to the work of Bhatt–Scholze, Scholze, and Wagner (cf. [BS22; Sch16; Wag25]). The following result will be used throughout.

Lemma 2.1 (Derived Nakayama)**.** — *Let I be a finitely generated ideal of a ring A. Let K be a derived complete object of the derived category $D(A)$. Then $K = 0$ if and only if $K \otimes^L_A A/I = 0$.*

Proof. This is [Sta, Tag 0G1U]. □

Theorem 2.2 (Bhatt–Scholze, Wagner)**.** — *There is a functor*

$$q\Omega_{-/\mathbf{Z}} \colon \mathrm{CAlg}^{\mathrm{sm}}_{\mathbf{Z}} \longrightarrow \mathrm{CAlg}(D(\mathbf{Z}[\![q-1]\!])^\wedge_{(q-1)})$$

from smooth $\mathbf{Z}$*-algebras to* $\mathbf{E}_\infty$*-algebras in* $D(\mathbf{Z}[\![q-1]\!])^\wedge_{(q-1)}$*, together with an equivalence*

$$q\Omega_{-/\mathbf{Z}} \otimes^L_{\mathbf{Z}[\![q-1]\!]} \mathbf{Z} \simeq \Omega^\bullet_{-/\mathbf{Z}}$$

of functors to $\mathrm{CAlg}(D(\mathbf{Z}))$.

Proof. For all but the last statement, this is [Wag25, Theorem 1.5]. It remains to see that the equivalence can be chosen as an equivalence of functors to $\mathrm{CAlg}(D(\mathbf{Z}))$. We make the preliminary observation that

$$-/(q-1) = - \otimes^L_{\mathbf{Z}[\![q-1]\!]} \mathbf{Z} \colon D(\mathbf{Z}[\![q-1]\!])^\wedge_{(q-1)} \longrightarrow D(\mathbf{Z})$$

is exact and symmetric monoidal. Hence the induced functor on $\mathbf{E}_\infty$-algebras preserves finite limits by [Lur17, Corollary 3.2.2.5]. Recall that the (global) q-de Rham complex is defined by the pullback square

$$\begin{array}{ccc} q\Omega_{R/\mathbf{Z}} & \longrightarrow & \prod_p q\Omega^\wedge_{R,p} \\ \downarrow & & \downarrow \\ (\Omega^\bullet_{R/\mathbf{Z}} \otimes^L_{\mathbf{Z}} \mathbf{Q})[\![q-1]\!] & \longrightarrow & \Big(\prod_p \Omega^{\bullet,\wedge}_{R,p} \otimes^L_{\mathbf{Z}} \mathbf{Q}\Big)[\![q-1]\!], \end{array}$$

of $\mathbf{E}_\infty$-algebras, where $q\Omega^\wedge_{R,p}$ is the q-crystalline cohomology of [BS22]. Applying $-/(q-1)$ yields another pullback square of functors to $\mathrm{CAlg}(D(\mathbf{Z}))$, and it simply remains to identify its terms. The bottom row becomes the canonical map $\Omega^\bullet_R \otimes^L_{\mathbf{Z}} \mathbf{Q} \longrightarrow \prod_p \Omega^{\bullet,\wedge}_{R,p} \otimes^L_{\mathbf{Z}} \mathbf{Q}$. The top-right corner becomes $\prod_p \Omega^{\bullet,\wedge}_{R,p}$ by first base-changing along the map of q-PD pairs $(\mathbf{Z}_p[\![q-1]\!], (q-1)) \longrightarrow (\mathbf{Z}_p, (p))$ and then applying the crystalline-de Rham comparison. Hence, $q\Omega_{R/\mathbf{Z}}/(q-1)$ is naturally equivalent, as an $\mathbf{E}_\infty$-algebra, to the pullback of

$$\Omega^\bullet_R \otimes^L_{\mathbf{Z}} \mathbf{Q} \longrightarrow \prod_p \Omega^{\bullet,\wedge}_{R,p} \otimes^L_{\mathbf{Z}} \mathbf{Q} \longleftarrow \prod_p \Omega^{\bullet,\wedge}_{R,p}.$$

By the arithmetic fracture square, this pullback is precisely $\Omega^\bullet_R$, which is a pullback of $\mathbf{E}_\infty$-algebras because the forgetful functor $\mathrm{CAlg}(D(\mathbf{Z})) \longrightarrow D(\mathbf{Z})$ creates limits. □

Proposition 2.3. — *The right Kan extension*

$$q\text{-dR} \colon \mathrm{Sm}^{\mathrm{op}}_{\mathbf{Z}} \longrightarrow D(\mathbf{Z}[\![q-1]\!])^\wedge_{(q-1)}$$

of $q\Omega_{-/\mathbf{Z}}$ along the inclusion $\mathrm{CAlg}^{\mathrm{sm}}_{\mathbf{Z}} \subseteq \mathrm{Sm}^{\mathrm{op}}_{\mathbf{Z}}$ satisfies étale descent. In particular, it is a Nisnevich sheaf. Moreover, there is a natural equivalence

$$q\text{-dR}(X)/(q-1) \simeq R\Gamma(X, \Omega^\bullet_{X/\mathbf{Z}}),$$

for any smooth $\mathbf{Z}$-scheme X.

Proof. To show that q-dR satisfies étale descent, it will suffice to show that $q\Omega_{-/\mathbf{Z}}$ does so on smooth affine schemes. To prove the latter, it will further suffice, by [Lur18, Proposition A.3.3.1], to show two things. First, that $q\Omega_{-/\mathbf{Z}}$ sends finite products of rings to products. Second, that for every faithfully flat étale map $R \longrightarrow R'$ of smooth $\mathbf{Z}$-algebras, the map

$$q\Omega_{R/\mathbf{Z}} \longrightarrow \lim_{[n]\in\Delta} q\Omega_{R'^{\otimes(n+1)}/\mathbf{Z}}$$

is an equivalence. In both cases, the fiber of the map in question is a limit of derived $(q-1)$-complete objects, and hence is derived $(q-1)$-complete. So by the derived Nakayama lemma it suffices to show that the fiber vanishes modulo $(q-1)$. Since $-/(q-1)$ commutes with limits, Theorem 2.2 reduces both claims to the corresponding statements for the de Rham complex $\Omega^\bullet_{-/\mathbf{Z}}$.

For finite products this is clear. For the second claim, note that $\Omega^1_{R/\mathbf{Z}}$ is a finite projective R-module, and let d be the maximum of its ranks. Since $\Omega^i_{R'^{\otimes(n+1)}/\mathbf{Z}} \cong \Omega^i_{R/\mathbf{Z}} \otimes_R R'^{\otimes(n+1)}$, all these modules vanish for $i > d$, so the stupid filtrations are finite of length d, uniformly

in n, so it suffices to treat each $\Omega^i_{-/\mathbf{Z}}$ separately for $0 \leq i \leq d$. In this case the claim follows from faithfully flat descent for modules [Sta, Tag 023M].

Finally, we treat the last claim in the statement of the proposition. Since $-/(q-1)$ commutes with limits, q-dR$(-)/(q-1)$ is the right Kan extension of $q\Omega_{-/\mathbf{Z}}/(q-1) \simeq \Omega^\bullet_{-/\mathbf{Z}}$, where we use Theorem 2.2. On the other hand, $R\Gamma(-, \Omega^\bullet_{-/\mathbf{Z}})$ is a Zariski sheaf on $\mathrm{Sm}_{\mathbf{Z}}$, and its restriction to affines is precisely $\Omega^\bullet_{-/\mathbf{Z}}$. It is therefore the right Kan extension of $\Omega^\bullet_{-/\mathbf{Z}}$, finishing the proof. □

Lemma 2.4. — *The sheaf q-dR is an $\mathbf{E}_\infty$-algebra in $\mathrm{Sh}_{\mathrm{Nis}}(\mathrm{Sm}_{\mathbf{Z}}, D(\mathbf{Z}[\![q-1]\!])^\wedge_{(q-1)})$ for the localized Day convolution symmetric monoidal structure. In other words, it is lax symmetric monoidal as a functor.*

Proof. Nisnevich localization is compatible with Day convolution since Nisnevich coverings are stable under products with a fixed scheme. So $q\Omega_{-/\mathbf{Z}}$, which is a sheaf by Proposition 2.3, is an $\mathbf{E}_\infty$-algebra in $\mathrm{Sh}_{\mathrm{Nis}}(\mathrm{Sm}^{\mathrm{aff}}_{\mathbf{Z}}, D(\mathbf{Z}[\![q-1]\!])^\wedge_{(q-1)})$. We apply [Hoy14, Lemma C.3] twice (the intermediate step being separated schemes) to get an equivalence $\mathrm{Sh}_{\mathrm{Nis}}(\mathrm{Sm}_{\mathbf{Z}}) \simeq \mathrm{Sh}_{\mathrm{Nis}}(\mathrm{CAlg}^{\mathrm{sm,op}}_{\mathbf{Z}})$ given by restriction, with inverse right Kan extension, which induces the same equivalence with values in $D(\mathbf{Z}[\![q-1]\!])^\wedge_{(q-1)}$ by taking the Lurie tensor product. This equivalence is symmetric monoidal for the localized Day convolutions, and q-dR corresponds to $q\Omega_{-/\mathbf{Z}}$ under it. □

Proposition 2.5. — *Let X and Y be smooth schemes over $\mathbf{Z}$. Then the canonical map*

$$q\text{-dR}(X) \mathbin{\widehat{\otimes}^L_{\mathbf{Z}[\![q-1]\!]}} q\text{-dR}(Y) \longrightarrow q\text{-dR}(X \times Y)$$

obtained from the lax symmetric monoidal structure on q-dR is an equivalence. In particular, the lax symmetric monoidal structure on q-dR is symmetric monoidal.

Proof. By the derived Nakayama lemma we may check this after reducing mod $(q-1)$, whence we need to show that the natural map

$$R\Gamma(X, \Omega^\bullet_X) \otimes^L_{\mathbf{Z}} R\Gamma(Y, \Omega^\bullet_Y) \longrightarrow R\Gamma(X \times Y, \Omega^\bullet_{X\times Y})$$

is an equivalence. The usual induction argument involving quasi-compact and quasi-separated schemes[1] and a Mayer–Vietoris sequence allows us to reduce to the case when X and Y are affine. In this case the result follows from [Sta, Tag 0FMB]. To see that q-dR actually is symmetric monoidal, we still need to verify that the natural map

$$\mathbf{Z}[\![q-1]\!] \longrightarrow q\text{-dR}(\mathbf{Z})$$

is an equivalence. But this can once again be checked mod $(q-1)$, in which case it is clear. □

Lemma 2.6. — *Let us write $\widetilde{q\text{-dR}}(\mathbf{P}^1) = \mathrm{fib}(q\text{-dR}(\mathbf{P}^1) \xrightarrow{\infty^*} q\text{-dR}(\mathbf{Z}))$ for the reduced q-de Rham cohomology of the projective line, and put $\mathbf{L} := \widetilde{H}^2_{q\text{-dR}}(\mathbf{P}^1) = H^2(\widetilde{q\text{-dR}}(\mathbf{P}^1))$. Then $\mathbf{L}$ is an invertible $\mathbf{Z}[\![q-1]\!]$-module, and we have*

$$\widetilde{q\text{-dR}}(\mathbf{P}^1) \simeq \mathbf{L}[-2], \qquad q\text{-dR}(\mathbf{P}^1) \simeq \mathbf{Z}[\![q-1]\!] \oplus \mathbf{L}[-2].$$

Proof. Recall that all complexes and modules appearing here are derived $(q-1)$-complete. Furthermore, the fiber sequence

$$\widetilde{q\text{-dR}}(\mathbf{P}^1) \longrightarrow q\text{-dR}(\mathbf{P}^1) \xrightarrow{\infty^*} q\text{-dR}(\mathbf{Z})$$

[1] Recall that all our schemes are qcqs.

splits canonically in the stable category $D(\mathbf{Z}[\![q-1]\!])^{\wedge}_{(q-1)}$, yielding an equivalence

$$q\text{-dR}(\mathbf{P}^1) \simeq q\text{-dR}(\mathbf{Z}) \oplus \widetilde{q\text{-dR}}(\mathbf{P}^1) \simeq \mathbf{Z}[\![q-1]\!] \oplus \widetilde{q\text{-dR}}(\mathbf{P}^1)$$

using Proposition 2.5. Since $- \otimes^L_{\mathbf{Z}[\![q-1]\!]} \mathbf{Z}$ is exact, we get that

$$\widetilde{q\text{-dR}}(\mathbf{P}^1)/(q-1) \simeq \mathrm{fib}(R\Gamma(\mathbf{P}^1, \Omega^\bullet_{\mathbf{P}^1}) \xrightarrow{\infty^*} R\Gamma(\operatorname{Spec} \mathbf{Z}, \Omega^\bullet_{\mathbf{Z}})).$$

Now recall (cf. [Sta, Tag 0FMG]) that $H^0_{\mathrm{dR}}(\mathbf{P}^1) \simeq H^2_{\mathrm{dR}}(\mathbf{P}^1) \simeq \mathbf{Z}$ and $H^i_{\mathrm{dR}}(\mathbf{P}^1) = 0$ otherwise. Comparing this with the fiber sequence above, we see that

$$H^i(\widetilde{q\text{-dR}}(\mathbf{P}^1)/(q-1)) \simeq \ker(H^i_{\mathrm{dR}}(\mathbf{P}^1) \xrightarrow{\infty^*} H^i_{\mathrm{dR}}(\operatorname{Spec} \mathbf{Z})) \simeq \begin{cases} \mathbf{Z}, & i = 2 \\ 0, & i \neq 2. \end{cases}$$

In particular, this forces $\widetilde{q\text{-dR}}(\mathbf{P}^1)/(q-1) \simeq \mathbf{Z}[-2]$. We now want to lift this to $\widetilde{q\text{-dR}}(\mathbf{P}^1)$. To this end, consider the fiber sequence

$$\widetilde{q\text{-dR}}(\mathbf{P}^1) \xrightarrow{q-1} \widetilde{q\text{-dR}}(\mathbf{P}^1) \longrightarrow \widetilde{q\text{-dR}}(\mathbf{P}^1)/(q-1) \simeq \mathbf{Z}[-2].$$

Looking at the associated long exact sequence in cohomology, we see that $(q-1)$ is surjective on H^i for all $i \neq 2$. But by the derived Nakayama lemma, then $H^i(\widetilde{q\text{-dR}}(\mathbf{P}^1)) = 0$ for all $i \neq 2$, so that $\widetilde{q\text{-dR}}(\mathbf{P}^1) \simeq \mathbf{L}[-2]$ by a similar argument as above.

It remains to show that $\mathbf{L}$ is free of rank one. In degree 2, the long exact sequence from above reads

$$0 \longrightarrow \mathbf{L} \xrightarrow{q-1} \mathbf{L} \longrightarrow \mathbf{Z} \longrightarrow 0.$$

Choose an element $\ell \in \mathbf{L}$ mapping to a generator of $\mathbf{Z}$, and a map $f\colon \mathbf{Z}[\![q-1]\!] \longrightarrow \mathbf{L}$ with $f(1) = \ell$. Note that its cofiber is derived $(q-1)$-complete. By construction, $f/(q-1)$ is an equivalence,[2] so that $\mathrm{cofib}(f/(q-1)) \simeq 0$. By the derived Nakayama lemma, we conclude that $\mathrm{cofib}(f) \simeq 0$. This finishes the proof. □

Remark 2.7. — The choice of a generator $\ell \in \mathbf{L}$ lifting a fixed generator of $\mathbf{L}/(q-1)$ forms a torsor under $1 + (q-1)\mathbf{Z}[\![q-1]\!]$ and hence is certainly not unique. However, this will not be relevant for us.

3. How to represent a cohomology theory

Before getting into the actual results in Section 4, we recall the ambient setting we will work in, namely Annala–Hoyois–Iwasa's category MS of non-$\mathbf{A}^1$-invariant motivic spectra [AI23; AHI24a; AHI24b]. For this section we may work over an arbitrary base, so let S be any scheme. The goal of this section will be to explain how one might go about representing a cohomology theory of schemes in MS without referring to orientations.

Construction 3.1. — We say that a presheaf $F\colon \mathrm{Sm}_S^{\mathrm{op}} \longrightarrow \mathrm{Sp}$ satisfies *smooth blow-up excision* if $F(\emptyset) \simeq 0$, and F carries the blow-up square

$$\begin{array}{ccc} E & \longrightarrow & \mathrm{Bl}_Z(X) \\ \downarrow & & \downarrow \\ Z & \hookrightarrow & X \end{array}$$

to a pullback square for all closed immersions $Z \hookrightarrow X$ in Sm_S. We then write

$$\mathrm{Sh}_{\mathrm{Nis,sbe}}(\mathrm{Sm}_S, \mathrm{Sp}) \subseteq \mathrm{Sh}_{\mathrm{Nis}}(\mathrm{Sm}_S, \mathrm{Sp})$$

[2] Here we use that $q-1$ is injective on both $\mathbf{Z}[\![q-1]\!]$ and $\mathbf{L}$, whence the derived quotients agree with the usual quotients.

for the category of Nisnevich sheaves of spectra satisfying smooth blow-up excision. Mimicking the construction of the category of spectra in topology, we now want to invert $\mathbf{P}^1 = \mathbf{P}^1_S$ as follows. Denote by

$$\mathbf{T} = \mathrm{cofib}(\Sigma^\infty_+ S \xrightarrow{\infty} \Sigma^\infty_+ \mathbf{P}^1) \in \mathrm{Sh}_{\mathrm{Nis,sbe}}(\mathrm{Sm}_S, \mathrm{Sp})$$

the sheaf represented by "$\mathbf{P}^1$ pointed at ∞". We then define

$$\mathrm{MS}_S := \mathrm{Sh}_{\mathrm{Nis,sbe}}(\mathrm{Sm}_S, \mathrm{Sp})[\mathbf{T}^{-1}]$$

and call it the category of *motivic spectra*. When $S = \operatorname{Spec} R$ is affine we simply write MS_R.

Remark 3.2. — We note that the inversion of $\mathbf{T}$ above is *not* the naive telescope, but rather Robalo's construction [Rob15]. We will explore the relation between the two notions below. See also [CF26] and [AI23] for a concrete description of MS_S in terms of symmetric $\mathbf{T}$-spectra.

Observe that MS_S is presentably symmetric monoidal and stable, and comes equipped with a natural symmetric monoidal functor

$$\mathrm{Sm}_S \xrightarrow{\Sigma^\infty_+} \mathrm{Sh}_{\mathrm{Nis,sbe}}(\mathrm{Sm}_S, \mathrm{Sp}) \longrightarrow \mathrm{MS}_S,$$

which we may denote by $\Sigma^\infty_{\mathbf{T}}$ or $(-)_+$. Moreover, the category MS_S enjoys the following universal property. Let $\mathcal{C}$ be a stable presentably symmetric monoidal category with a symmetric monoidal functor $\Phi\colon \mathrm{Sm}_S \longrightarrow \mathcal{C}$ such that Φ sends Nisnevich and smooth blow-up squares to pullback squares, and $\mathrm{cofib}(\Phi(S) \xrightarrow{\infty} \Phi(\mathbf{P}^1))$ is invertible in $\mathcal{C}$. Then there is a unique symmetric monoidal left adjoint $\mathrm{MS}_S \longrightarrow \mathcal{C}$ filling the diagram

$$\begin{array}{ccc} \mathrm{Sm}_S & \xrightarrow{\Sigma^\infty_{\mathbf{T}}} & \mathrm{MS}_S \\ & \searrow^{\Phi} & \downarrow \\ & & \mathcal{C}. \end{array}$$

We also note that, essentially by construction, there is an adjoint pair of autoequivalences

$$\Sigma_{\mathbf{T}}\colon \mathrm{MS}_S \rightleftarrows \mathrm{MS}_S\colon \Omega_{\mathbf{T}},$$

where we may also write $\Sigma_{\mathbf{T}}(-) = - \otimes \mathbf{T}$ and $\Omega_{\mathbf{T}}(-) = \underline{\mathrm{Hom}}(\mathbf{T}, -)$. With a slight risk of confusion, we will use the same notation for the functors on the level of $\mathrm{Sh}_{\mathrm{Nis,sbe}}(\mathrm{Sm}_S, \mathrm{Sp})$. In particular, we do not distinguish between $\mathbf{T}$ viewed as an object of $\mathrm{Sh}_{\mathrm{Nis,sbe}}(\mathrm{Sm}_S, \mathrm{Sp})$ and as an object of MS_S in the notation.

Definition 3.3. — For $E \in \mathrm{MS}_S$ and $n \in \mathbf{Z}$, we put

$$E_n := \Omega^\infty_{\mathbf{T}} \Sigma^n_{\mathbf{T}} E \in \mathrm{Sh}_{\mathrm{Nis,sbe}}(\mathrm{Sm}_S, \mathrm{Sp}),$$

where $\Omega^\infty_{\mathbf{T}}\colon \mathrm{MS}_S \longrightarrow \mathrm{Sh}_{\mathrm{Nis,sbe}}(\mathrm{Sm}_S, \mathrm{Sp})$ is the right adjoint of $\Sigma^\infty_{\mathbf{T}}$. We call $(E_n)_{n\geq 0}$ the underlying sequential spectrum of E.

Lemma 3.4. — *Let X and Y be smooth S-schemes, and let $F \in \mathrm{Sh}_{\mathrm{Nis,sbe}}(\mathrm{Sm}_S, \mathrm{Sp})$.*

(1) There is a natural equivalence

$$\underline{\mathrm{Hom}}(\Sigma^\infty_+ X, F)(Y) \simeq F(X \times Y).$$

In particular, there is a natural identification

$$\Omega_{\mathbf{T}}(F)(X) \simeq \mathrm{fib}(F(X \times \mathbf{P}^1) \xrightarrow{\infty^*} F(X)).$$

(2) *For $n \geq 1$, there is a natural equivalence $\Omega^n_{\mathbf{T}}(F) \simeq \underline{\mathrm{Hom}}(\mathbf{T}^{\otimes n}, F)$. Moreover, evaluating at X gives*

$$\Omega^n_{\mathbf{T}}(F)(X) \simeq \mathrm{tfib}_{I \subseteq [n]}(F(X \times (\mathbf{P}^1)^I)),$$

where we put $(\mathbf{P}^1)^\emptyset = S$, and for $J \subseteq I$ the map $F(X \times (\mathbf{P}^1)^I) \longrightarrow F(X \times (\mathbf{P}^1)^J)$ is restriction along the closed immersion $(\mathbf{P}^1)^J \hookrightarrow (\mathbf{P}^1)^I$ which is ∞ in the coordinates indexed by $I \smallsetminus J$. Under the above identification, the action of $\sigma \in \Sigma_n$ on $\Omega^n_{\mathbf{T}}(F)$ through the symmetric monoidal structure corresponds to the automorphism of the cube sending the vertex I to the vertex $\sigma(I)$. In particular, on the vertex $I = [n]$ it is given by pullback along the permutation of the factors of $(\mathbf{P}^1)^n$.

(3) *For $E \in \mathrm{MS}_S$ and $n \in \mathbf{Z}$, there are natural equivalences*

$$E_n(X) \simeq \mathrm{map}_{\mathrm{MS}_S}(X_+, \Sigma^n_{\mathbf{T}} E), \quad E_n \simeq \Omega_{\mathbf{T}} E_{n+1}, \quad (\Omega_{\mathbf{T}} E)_n \simeq \Omega_{\mathbf{T}}(E_n),$$

where in the last equivalence the first instance of $\Omega_{\mathbf{T}}$ is computed in MS_S and the latter one in $\mathrm{Sh}_{\mathrm{Nis,sbe}}(\mathrm{Sm}_S, \mathrm{Sp})$. In particular, we have

$$(\Omega_{\mathbf{T}} E)_n(X) \simeq \mathrm{fib}(E_n(X \times \mathbf{P}^1) \xrightarrow{\infty^*} E_n(X))$$

for all $n \in \mathbf{Z}$.

Proof. (1) For the first claim, this is just the Yoneda lemma combined with the fact that Σ^∞_+ is symmetric monoidal. For the latter, apply the exact functor $\underline{\mathrm{Hom}}(-, F)$ to the cofiber sequence

$$\Sigma^\infty_+ S \xrightarrow{\infty} \Sigma^\infty_+ \mathbf{P}^1 \longrightarrow \mathbf{T}$$

and evaluate at X.

(2) Iterating the tensor-hom adjunction we see that $\Omega^n_{\mathbf{T}}(F) \simeq \underline{\mathrm{Hom}}(\mathbf{T}^{\otimes n}, F)$. Next, note that we can write

$$\mathbf{T}^{\otimes n} \simeq \mathrm{tcofib}_{I \subseteq [n]}(\Sigma^\infty_+ (\mathbf{P}^1)^I).$$

Thus, applying the exact functor $\underline{\mathrm{Hom}}(-, F)$, evaluating at X, and applying (1) yields the desired identification. The identification of the Σ_n-action amounts to the fact that Σ^∞_+ is symmetric monoidal.

(3) For the first equivalence, use the adjunction between $\Sigma^\infty_{\mathbf{T}}$ and $\Omega^\infty_{\mathbf{T}}$ combined with the Yoneda lemma. Next, observe that for any $G \in \mathrm{Sh}_{\mathrm{Nis,sbe}}(\mathrm{Sm}_S, \mathrm{Sp})$, we have

$$\begin{aligned}
\mathrm{map}(G, \Omega^\infty_{\mathbf{T}} \underline{\mathrm{Hom}}(\mathbf{T}, E)) &\simeq \mathrm{map}(\Sigma^\infty_{\mathbf{T}} G, \underline{\mathrm{Hom}}(\mathbf{T}, E)) \\
&\simeq \mathrm{map}(\Sigma^\infty_{\mathbf{T}} G \otimes \mathbf{T}, E) \\
&\simeq \mathrm{map}(\Sigma^\infty_{\mathbf{T}} (G \otimes \mathbf{T}), E) \\
&\simeq \mathrm{map}(G \otimes \mathbf{T}, \Omega^\infty_{\mathbf{T}} E) \\
&\simeq \mathrm{map}(G, \underline{\mathrm{Hom}}(\mathbf{T}, \Omega^\infty_{\mathbf{T}} E)).
\end{aligned}$$

This shows that

$$\Omega^\infty_{\mathbf{T}} \Omega_{\mathbf{T}} E = \Omega^\infty_{\mathbf{T}} \underline{\mathrm{Hom}}(\mathbf{T}, E) \simeq \underline{\mathrm{Hom}}(\mathbf{T}, \Omega^\infty_{\mathbf{T}} E) = \Omega_{\mathbf{T}} \Omega^\infty_{\mathbf{T}} E,$$

which easily yields the last two equivalences. □

Although it is hard to get a grasp of objects in MS_S, the following two statements show that we may often reduce to checking statements on the level of telescopes. Here we write

$$\mathrm{Tel}_X(\mathcal{C}) := \mathrm{colim}(\mathcal{C} \xrightarrow{-\otimes X} \mathcal{C} \xrightarrow{-\otimes X} \cdots)$$

for an object X in a presentably symmetric monoidal stable category $\mathcal{C}$, the colimit being taken in Pr^L. Since colimits in Pr^L are computed as limits in Pr^R along the right adjoints, an object of $\mathrm{Tel}_X(\mathcal{C})$ is a sequence $(Y_n)_{n \geq 0}$ of objects of $\mathcal{C}$ together with equivalences $Y_n \simeq \underline{\mathrm{Hom}}(X, Y_{n+1})$, which are part of the data.

Proposition 3.5 ([Hoy26]). — *Let $\mathcal{C}$ be a stable presentably symmetric monoidal category, and let $X \in \mathcal{C}$ be an object. Then the forgetful functor $\mathrm{Sp}_X(\mathcal{C}) \longrightarrow \mathrm{Tel}_X(\mathcal{C})$ is fully faithful. Its essential image is $\mathrm{Tel}_X(\mathcal{C}')$, where $\mathcal{C}' \subseteq \mathcal{C}$ is the full subcategory spanned by objects Y such that the cyclic permutation of $\underline{\mathrm{Hom}}(X^{\otimes 3}, Y)$ is homotopic to the identity.*

Proposition 3.6. — *Let $(E_n)_{n\geq 0}$ be a sequence of objects in $\mathrm{Sh}_{\mathrm{Nis,sbe}}(\mathrm{Sm}_S, \mathrm{Sp})$ together with equivalences $E_n \simeq \Omega_{\mathbf{T}} E_{n+1}$, and such that the cyclic permutation of the factors of $\mathbf{T}^{\otimes 3}$ acts on $\Omega^3_{\mathbf{T}} E_n$ by maps homotopic to the identity for all $n \geq 0$. Then there is a unique $E \in \mathrm{MS}_S$ with underlying sequential spectrum $(E_n)_{n\geq 0}$.*

Proof. Apply Proposition 3.5 with $X = \mathbf{T}$. □

Remark 3.7. — As alluded to at the beginning of the section, one can bypass checking smooth blow-up excision if it is known that the projective bundle formula holds. Indeed, this is equivalent to elementary blow-up excision by [AI23, Lemma 3.3.5], which in turn is equivalent to smooth blow-up excision by [AHI24a, Proposition 2.2] as we are working with Nisnevich sheaves. However, this requires a theory of Chern classes, which we do not pursue here.

4. Representability of q-de Rham cohomology

In view of the results from Section 2, we really only have one way to define the candidate for the motivic spectrum representing q-de Rham cohomology. We now specialize to $S = \operatorname{Spec} \mathbf{Z}$, and view objects of $D(\mathbf{Z}[\![q-1]\!])^{\wedge}_{(q-1)}$ as spectra via the forgetful functor. This functor preserves all limits, hence sheaves and blow-up excision.

For $n \in \mathbf{Z}$, consider the Nisnevich sheaves of spectra

$$q\text{-}\mathrm{dR}^{\mathrm{mot}}_n \colon X \longmapsto q\text{-}\mathrm{dR}(X) \otimes_{\mathbf{Z}[\![q-1]\!]} \mathbf{L}^{\otimes -n}[2n] \in \mathrm{Sh}_{\mathrm{Nis}}(\mathrm{Sm}_{\mathbf{Z}}, \mathrm{Sp}).$$

Since $\mathbf{L}$ is free of rank one, the tensor product is underived and agrees with the completed one. Our next goal will be to show that these sheaves satisfy smooth blow-up excision and assemble to a sequential spectrum.

Proposition 4.1. — *The sheaf q-dR satisfies smooth blow-up excision.*

Proof. Let $Z \hookrightarrow X$ be a closed immersion of smooth $\mathbf{Z}$-schemes, with blow-up X' and exceptional divisor E. Consider the resulting diagram

$$\begin{array}{ccc} F & \dashrightarrow & F' \\ \downarrow & & \downarrow \\ q\text{-}\mathrm{dR}(X) & \longrightarrow & q\text{-}\mathrm{dR}(X') \\ \downarrow & & \downarrow \\ q\text{-}\mathrm{dR}(Z) & \longrightarrow & q\text{-}\mathrm{dR}(E), \end{array}$$

where F and F' denote the fibers of the corresponding maps. To show that the lower square is cartesian, it will suffice to show that the induced map between the fibers is an equivalence. By the derived Nakayama lemma, this can be checked after reducing mod $(q-1)$. In this case it is known to be true, for example, as de Rham cohomology over $\mathbf{Z}$ is representable in $\mathrm{MS}_{\mathbf{Z}}$. Alternatively, use the stupid filtration to reduce to Hodge cohomology and apply [Sta, Tag 0FUC]. □

Corollary 4.2. — *The sheaves q-$\mathrm{dR}^{\mathrm{mot}}_n$ satisfy smooth blow-up excision for all $n \in \mathbf{Z}$.*

Proof. Fix an integer n. The functor $- \otimes^L_{\mathbf{Z}[\![q-1]\!]} \mathbf{L}^{\otimes -n}[2n]$ is an exact autoequivalence of $D(\mathbf{Z}[\![q-1]\!])^\wedge_{(q-1)}$ as $\mathbf{L}$ is invertible. In particular, it preserves pullback squares, as does the forgetful functor to spectra. The claim then follows from Proposition 4.1. □

Proposition 4.3. — *For each $n \in \mathbf{Z}$, there is a natural equivalence*

$$\sigma_n \colon q\text{-dR}^{\mathrm{mot}}_n \xrightarrow{\simeq} \Omega_{\mathbf{T}}\, q\text{-dR}^{\mathrm{mot}}_{n+1}$$

in $\mathrm{Sh}_{\mathrm{Nis,sbe}}(\mathrm{Sm}_{\mathbf{Z}}, \mathrm{Sp})$.

Proof. By Lemma 3.4(1) we have, naturally in X,

$$\Omega_{\mathbf{T}}\, q\text{-dR}^{\mathrm{mot}}_{n+1}(X) \simeq \mathrm{fib}\left(q\text{-dR}(X \times \mathbf{P}^1) \xrightarrow{\infty^*} q\text{-dR}(X)\right) \otimes \mathbf{L}^{\otimes -n-1}[2n+2].$$

The external products of q-dR are natural in both variables, so the section $\infty \colon \operatorname{Spec} \mathbf{Z} \longrightarrow \mathbf{P}^1$ induces a commutative square of functors in X

$$\begin{array}{ccc} q\text{-dR}(X) \mathbin{\widehat{\otimes}^L_{\mathbf{Z}[\![q-1]\!]}} q\text{-dR}(\mathbf{P}^1) & \xrightarrow{\simeq} & q\text{-dR}(X \times \mathbf{P}^1) \\ \downarrow{\scriptstyle \mathrm{id} \otimes \infty^*} & & \downarrow{\scriptstyle \infty^*} \\ q\text{-dR}(X) \mathbin{\widehat{\otimes}^L_{\mathbf{Z}[\![q-1]\!]}} q\text{-dR}(\mathbf{Z}) & \xrightarrow{\simeq} & q\text{-dR}(X), \end{array}$$

whose horizontal maps are equivalences by Proposition 2.5. Taking fibers of the vertical maps, and using exactness of $q\text{-dR}(X) \mathbin{\widehat{\otimes}^L_{\mathbf{Z}[\![q-1]\!]}} -$ together with Lemma 2.6, we get a natural equivalence

$$\mathrm{fib}\left(q\text{-dR}(X \times \mathbf{P}^1) \xrightarrow{\infty^*} q\text{-dR}(X)\right) \simeq q\text{-dR}(X) \mathbin{\widehat{\otimes}^L_{\mathbf{Z}[\![q-1]\!]}} \widetilde{q\text{-dR}}(\mathbf{P}^1) \simeq q\text{-dR}(X) \otimes \mathbf{L}[-2].$$

Thus, using the canonical equivalence $\mathbf{L} \otimes \mathbf{L}^{\otimes -n-1} \simeq \mathbf{L}^{\otimes -n}$,

$$\begin{aligned} \Omega_{\mathbf{T}}\, q\text{-dR}^{\mathrm{mot}}_{n+1}(X) &\simeq q\text{-dR}(X) \otimes \mathbf{L}[-2] \otimes \mathbf{L}^{\otimes -n-1}[2n+2] \\ &\simeq q\text{-dR}(X) \otimes \mathbf{L}^{\otimes -n}[2n] \\ &= q\text{-dR}^{\mathrm{mot}}_n(X), \end{aligned}$$

naturally in X. We let σ_n be the inverse of this equivalence. □

Remark 4.4. — Any family of equivalences $q\text{-dR}^{\mathrm{mot}}_n \simeq \Omega_{\mathbf{T}}\, q\text{-dR}^{\mathrm{mot}}_{n+1}$ defines an object of $\mathrm{Tel}_{\mathbf{T}}(\mathrm{Sh}_{\mathrm{Nis,sbe}}(\mathrm{Sm}_{\mathbf{Z}}, \mathrm{Sp}))$, and different families may a priori give non-equivalent motivic spectra. Our σ_n involve no choices (in particular no generator of $\mathbf{L}$), so maps compatible with external products, such as reduction modulo $(q-1)$, commute with the σ_n and hence induce maps of motivic spectra by Proposition 3.5.

Lemma 4.5 ([Dug14])**.** — *Let X be an invertible object of a symmetric monoidal category $\mathcal{C}$. Then there is $\varepsilon_X \in \pi_0 \operatorname{Aut}(\mathbf{1})$ with $\varepsilon_X^2 = 1$ and such that every permutation $\sigma \in \Sigma_n$ acts on $X^{\otimes n}$ (up to homotopy) by ε_X if σ is odd and by the identity if σ is even. In particular, every 3-cycle acts on $X^{\otimes 3}$ by a map homotopic to the identity.*

Proof. Recall that for any invertible object $Y \in \mathcal{C}$, the map $- \otimes \mathrm{id}_Y \colon \pi_0 \operatorname{End}(\mathbf{1}) \longrightarrow \pi_0 \operatorname{End}(Y)$ is an isomorphism. In particular, there is a unique element $\varepsilon_X \in \pi_0 \operatorname{End}(\mathbf{1})$ corresponding to the swap $\tau \in \pi_0 \operatorname{End}(X \otimes X)$, and it satisfies $\varepsilon_X^2 = 1$. The adjacent transposition $s_i = (i\ \ i{+}1) \in \Sigma_n$ acts on $X^{\otimes n}$ by

$$\mathrm{id}_{X^{\otimes(i-1)}} \otimes \tau \otimes \mathrm{id}_{X^{\otimes(n-i-1)}} = \varepsilon_X \otimes \mathrm{id}_{X^{\otimes n}}.$$

The adjacent transpositions generate Σ_n, and if $\sigma = s_{i_1} \cdots s_{i_\ell}$ then ℓ is even if and only if σ is even. Hence σ acts by $\varepsilon_X^\ell \otimes \mathrm{id}_{X^{\otimes n}}$, which is id for σ even and $\varepsilon_X \otimes \mathrm{id}$ for σ odd, because $\varepsilon_X^2 = 1$. □

Proposition 4.6. — *For each $n \in \mathbf{Z}$, the cyclic permutation of the factors of $\mathbf{T}^{\otimes 3}$ acts on $\Omega^3_{\mathbf{T}}\, q\text{-dR}^{\mathrm{mot}}_n$ by a map homotopic to the identity.*

Proof. By Lemma 3.4(2),

$$\Omega^3_{\mathbf{T}}\, q\text{-dR}^{\mathrm{mot}}_n(X) \simeq \mathrm{tfib}_{I \subseteq [3]}\big(\, q\text{-dR}^{\mathrm{mot}}_n(X \times (\mathbf{P}^1)^I)\big),$$

and the cyclic permutation σ acts through the automorphism of the cube sending the vertex I to $\sigma(I)$ by relabeling coordinates. Since q-dR is symmetric monoidal, any identifications are Σ_3-equivariant. In particular, one unravels the notation to see that

$$\Omega^3_{\mathbf{T}}\, q\text{-dR}^{\mathrm{mot}}_n(X) \simeq q\text{-dR}^{\mathrm{mot}}_n(X) \otimes (\mathbf{L}[-2])^{\otimes 3}$$

naturally in X. Under this equivalence, σ acts by $\mathrm{id} \otimes \gamma$, where γ is the action of σ on $(\mathbf{L}[-2])^{\otimes 3}$ through the symmetric monoidal structure of $D(\mathbf{Z}[\![q-1]\!])^{\wedge}_{(q-1)}$. Since $\mathbf{L}[-2]$ is invertible, Lemma 4.5 shows that γ is homotopic to the identity, σ being even. Finally, it is easy to see that this homotopy can be chosen naturally in X, i.e., that it is a homotopy in $\mathrm{Sh}_{\mathrm{Nis,sbe}}(\mathrm{Sm}_{\mathbf{Z}}, \mathrm{Sp})$. This proves the claim. □

Theorem 4.7. — *There is a motivic spectrum $q\text{-dR}^{\mathrm{mot}} \in \mathrm{MS}_{\mathbf{Z}}$ with underlying sequential spectrum $(q\text{-dR}^{\mathrm{mot}}_n, \sigma_n)_{n \geq 0}$, which represents global q-de Rham cohomology in the sense that*

$$\pi_{2m-n} \mathrm{map}_{\mathrm{MS}_{\mathbf{Z}}}(X_+, \Sigma^m_{\mathbf{T}}\, q\text{-dR}^{\mathrm{mot}}) \cong H^n_{q\text{-dR}}(X) \otimes_{\mathbf{Z}[\![q-1]\!]} \mathbf{L}^{\otimes -m}$$

for all smooth $\mathbf{Z}$-schemes X and $n, m \in \mathbf{Z}$. In particular, taking $m = 0$ recovers untwisted global q-de Rham cohomology.

Proof. By Corollary 4.2, Proposition 4.3 and Proposition 4.6, the sequential spectrum $(q\text{-dR}^{\mathrm{mot}}_n, \sigma_n)_{n \geq 0}$ satisfies the hypotheses of Proposition 3.6. This gives the motivic spectrum $q\text{-dR}^{\mathrm{mot}} \in \mathrm{MS}_{\mathbf{Z}}$. By Lemma 3.4(3), together with Proposition 4.3 for negative m, we have $\mathrm{map}_{\mathrm{MS}_{\mathbf{Z}}}(X_+, \Sigma^m_{\mathbf{T}}\, q\text{-dR}^{\mathrm{mot}}) \simeq q\text{-dR}(X) \otimes \mathbf{L}^{\otimes -m}[2m]$. The formula follows from the fact that $\mathbf{L}^{\otimes -m}$ is free, and hence flat, so that

$$\begin{aligned} \pi_{2m-n} \mathrm{map}_{\mathrm{MS}_{\mathbf{Z}}}(X_+, \Sigma^m_{\mathbf{T}}\, q\text{-dR}^{\mathrm{mot}}) &\simeq \pi_{2m-n}(q\text{-dR}(X) \otimes \mathbf{L}^{\otimes -m}[2m]) \\ &\simeq H^n(q\text{-dR}(X)) \otimes \mathbf{L}^{\otimes -m} \\ &= H^n_{q\text{-dR}}(X) \otimes \mathbf{L}^{\otimes -m}. \end{aligned}$$

□

5. Relation to de Rham and prismatic cohomology

We now show that the motivic spectrum $q\text{-dR}^{\mathrm{mot}}$ supplied by Theorem 4.7 is compatible with the known relations of q-de Rham cohomology with de Rham and prismatic cohomology. More precisely, its reduction mod $(q-1)$ is de Rham cohomology, and its p-completion is relative prismatic cohomology over the q-de Rham prism. It will be convenient to add a slight level of complexity to our construction.

Lemma 5.1. — *Write $\mathrm{Fun}^{\mathrm{ex}}$ for the category of exact functors $D(\mathbf{Z}[\![q-1]\!])^{\wedge}_{(q-1)} \longrightarrow \mathrm{Sp}$. For $\Phi \in \mathrm{Fun}^{\mathrm{ex}}$ and $n \in \mathbf{Z}$, write $\Phi\, q\text{-dR}^{\mathrm{mot}}_n$ for the presheaf $X \mapsto \Phi(q\text{-dR}^{\mathrm{mot}}_n(X))$. Then the following hold.*

(1) Each $\Phi\, q\text{-dR}^{\mathrm{mot}}_n$ lies in $\mathrm{Sh}_{\mathrm{Nis,sbe}}(\mathrm{Sm}_{\mathbf{Z}}, \mathrm{Sp})$, and the maps $\Phi(\sigma_n)$ are equivalences $\Phi\, q\text{-dR}^{\mathrm{mot}}_n \simeq \Omega_{\mathbf{T}} \Phi\, q\text{-dR}^{\mathrm{mot}}_{n+1}$.

(2) *There is a functor* $\mathrm{Fun}^{\mathrm{ex}} \longrightarrow \mathrm{MS}_{\mathbf{Z}}$, $\Phi \mapsto \Phi\, q\text{-dR}^{\mathrm{mot}}$, *such that the underlying sequential spectrum of* $\Phi\, q\text{-dR}^{\mathrm{mot}}$ *is* $(\Phi\, q\text{-dR}^{\mathrm{mot}}_n, \Phi(\sigma_n))_{n\geq 0}$. *In particular,*

$$\mathrm{map}_{\mathrm{MS}_{\mathbf{Z}}}(X_+, \Sigma^m_{\mathbf{T}} \Phi\, q\text{-dR}^{\mathrm{mot}}) \simeq \Phi(q\text{-dR}(X) \otimes \mathbf{L}^{\otimes -m}[2m])$$

for all $X \in \mathrm{Sm}_{\mathbf{Z}}$ *and* $m \in \mathbf{Z}$.

(3) *This functor preserves limits.*

Proof. (1) Since Φ preserves finite limits and the zero object, $\Phi\, q\text{-dR}^{\mathrm{mot}}_n$ satisfies smooth blow-up excision and sends Nisnevich squares to cartesian squares, hence is a Nisnevich sheaf. Since Φ commutes with the fiber in Lemma 3.4(1), $\Phi(\sigma_n)$ is an equivalence.

(2) For each n, $\Phi \mapsto \Phi\, q\text{-dR}^{\mathrm{mot}}_n$ is a functor $\mathrm{Fun}^{\mathrm{ex}} \longrightarrow \mathrm{Sh}_{\mathrm{Nis,sbe}}(\mathrm{Sm}_{\mathbf{Z}}, \mathrm{Sp})$, and the equivalences $\Phi(\sigma_n)$ are natural in Φ. Since $\mathrm{Tel}_{\mathbf{T}}$ is a limit over a tower indexed by $\mathbb{N}$, this data defines a functor $\mathrm{Fun}^{\mathrm{ex}} \longrightarrow \mathrm{Tel}_{\mathbf{T}}(\mathrm{Sh}_{\mathrm{Nis,sbe}}(\mathrm{Sm}_{\mathbf{Z}}, \mathrm{Sp}))$. Since Φ commutes with $\Omega^3_{\mathbf{T}}$, applying Φ to the homotopy constructed in the proof of Proposition 4.6 shows that this functor lands in the essential image of the fully faithful functor of Proposition 3.5. Hence it lifts uniquely to $\mathrm{MS}_{\mathbf{Z}}$. The formula follows from Lemma 3.4(3).

(3) Let $U\colon \mathrm{MS}_{\mathbf{Z}} \longrightarrow \mathrm{Tel}_{\mathbf{T}}(\mathrm{Sh}_{\mathrm{Nis,sbe}}(\mathrm{Sm}_{\mathbf{Z}}, \mathrm{Sp}))$ be the forgetful functor. It is fully faithful, hence conservative, and it preserves limits, since each $E \mapsto E_n$ does. Limits in $\mathrm{Tel}_{\mathbf{T}}$ are computed levelwise, and limits of sheaves are computed objectwise. So the composite of our functor with U preserves limits. It follows that our functor preserves limits as well: for a diagram (Φ_k), the comparison map $(\lim_k \Phi_k)\, q\text{-dR}^{\mathrm{mot}} \longrightarrow \lim_k(\Phi_k\, q\text{-dR}^{\mathrm{mot}})$ is sent by U to an equivalence. □

Example 5.2. — Taking Φ to be the forgetful functor recovers $q\text{-dR}^{\mathrm{mot}}$. Multiplication by an element $\lambda \in \mathbf{Z}[\![q-1]\!]$ is a natural endomorphism of the forgetful functor, so by the functoriality in Lemma 5.1(2) it defines an endomorphism λ of $q\text{-dR}^{\mathrm{mot}}$.

We start with the easy case of de Rham cohomology. Recall from the proof of Lemma 2.6 that

$$\mathbf{L}_{\mathrm{dR}} := \mathbf{L}/(q-1)\mathbf{L} \cong \widetilde{H}^2_{\mathrm{dR}}(\mathbf{P}^1_{\mathbf{Z}})$$

is a free $\mathbf{Z}$-module of rank one.

Proposition 5.3. — *Let* $q\text{-dR}^{\mathrm{mot}}/(q-1) := \mathrm{cofib}(q-1\colon q\text{-dR}^{\mathrm{mot}} \longrightarrow q\text{-dR}^{\mathrm{mot}})$. *For all* $X \in \mathrm{Sm}_{\mathbf{Z}}$ *and* $n, m \in \mathbf{Z}$, *we have*

$$\mathrm{map}_{\mathrm{MS}_{\mathbf{Z}}}(X_+, \Sigma^m_{\mathbf{T}}\, q\text{-dR}^{\mathrm{mot}}/(q-1)) \simeq R\Gamma(X, \Omega^\bullet_{X/\mathbf{Z}}) \otimes_{\mathbf{Z}} \mathbf{L}^{\otimes -m}_{\mathrm{dR}}[2m].$$

In particular,

$$\pi_{2m-n}\, \mathrm{map}_{\mathrm{MS}_{\mathbf{Z}}}(X_+, \Sigma^m_{\mathbf{T}}\, q\text{-dR}^{\mathrm{mot}}/(q-1)) \cong H^n_{\mathrm{dR}}(X) \otimes_{\mathbf{Z}} \mathbf{L}^{\otimes -m}_{\mathrm{dR}}.$$

Proof. Let Φ be the composition of $-/(q-1)$ followed by the forgetful functor. The cofiber sequence of exact functors $\mathrm{id} \xrightarrow{q-1} \mathrm{id} \longrightarrow \Phi$ is objectwise a cofiber sequence, so by Lemma 5.1(3) we get $\Phi\, q\text{-dR}^{\mathrm{mot}} \simeq q\text{-dR}^{\mathrm{mot}}/(q-1)$. By Lemma 5.1(2) and Proposition 2.3,

$$\begin{aligned}\mathrm{map}(X_+, \Sigma^m_{\mathbf{T}}\, q\text{-dR}^{\mathrm{mot}}/(q-1)) &\simeq (q\text{-dR}(X) \otimes \mathbf{L}^{\otimes -m}[2m])/(q-1)\\ &\simeq R\Gamma(X, \Omega^\bullet_{X/\mathbf{Z}}) \otimes_{\mathbf{Z}} \mathbf{L}^{\otimes -m}_{\mathrm{dR}}[2m].\end{aligned}$$

The second formula follows since $\mathbf{L}_{\mathrm{dR}}$ is free. □

Remark 5.4. — Annala–Pstrągowski construct a motivic spectrum $\mathrm{dR}(\mathbf{Z}) \in \mathrm{MS}_{\mathbf{Z}}$ representing de Rham cohomology, as an $\mathbf{E}_\infty$-H$\mathbf{Z}$-algebra with a Hodge-filtered refinement, using the de Rham orientation [AP26, Section 3]. Its bonding maps are multiplication by $c^{\mathrm{dR}}_1(\mathscr{O}(1))$, which therefore generates $\mathbf{L}_{\mathrm{dR}}$. Identifying $\mathbf{L}_{\mathrm{dR}} \cong \mathbf{Z}$ via this class, the levels of

q-$\mathrm{dR}^{\mathrm{mot}}/(q-1)$ and $\mathrm{dR}(\mathbf{Z})$ agree. Both families of bonding maps are external products with a generator of $\widetilde{H}^2_{\mathrm{dR}}(\mathbf{P}^1_{\mathbf{Z}})$; for q-$\mathrm{dR}^{\mathrm{mot}}/(q-1)$ this uses that the equivalence of Theorem 2.2 is multiplicative. So they agree up to sign, and as in the proof of Proposition 5.7 we get q-$\mathrm{dR}^{\mathrm{mot}}/(q-1) \simeq \mathrm{dR}(\mathbf{Z})$ by Proposition 3.5. In other words, q-$\mathrm{dR}^{\mathrm{mot}}$ is a q-deformation of their de Rham spectrum, constructed without orientations.

We now turn our attention to prismatic cohomology. Fix a prime p, and let $(A, I) = (\mathbf{Z}_p[\![q-1]\!], ([p]_q))$ be the q-de Rham prism. In particular, $A/I \cong \mathbf{Z}_p[\zeta_p]$. For $X \in \mathrm{Sm}_{\mathbf{Z}}$, let $\mathfrak{X}_p := (X \times_{\mathbf{Z}} \operatorname{Spec} \mathbf{Z}_p[\zeta_p])^\wedge_p$, which is a smooth p-adic formal scheme over A/I. We write $R\Gamma_{\mathbb{\Delta}}(\mathfrak{X}_p/A)$ for its relative prismatic cohomology [BS22].

Theorem 5.5 (Bhatt–Scholze, Wagner)**.** — *There is an equivalence of $\mathbf{E}_\infty$-algebras*

$$q\text{-}\mathrm{dR}(X)^\wedge_p \simeq R\Gamma_{\mathbb{\Delta}}(\mathfrak{X}_p/A),$$

natural in $X \in \mathrm{Sm}_{\mathbf{Z}}$.

Proof. For affine X this is [BS22, Section 16]. The equivalence is multiplicative by the same argument as in the proof of Theorem 2.2: by construction, q-$\mathrm{dR}(X)^\wedge_p$ is q-crystalline cohomology, and the comparison between q-crystalline and prismatic cohomology is induced by a map of sites. Both sides are Zariski sheaves in X, so the general case follows from the affine one, again by [Hoy14, Lemma C.3]. □

Proposition 5.6. — *Let $(q\text{-}\mathrm{dR}^{\mathrm{mot}})^\wedge_p := \lim_k q\text{-}\mathrm{dR}^{\mathrm{mot}}/p^k$ be the p-completion in $\mathrm{MS}_{\mathbf{Z}}$. Then $\mathbf{L}^\wedge_p \cong \widetilde{H}^2_{\mathbb{\Delta}}(\mathbf{P}^1_{A/I}/A)$ is a free A-module of rank one, and for all $X \in \mathrm{Sm}_{\mathbf{Z}}$ and $m \in \mathbf{Z}$,*

$$\mathrm{map}_{\mathrm{MS}_{\mathbf{Z}}}(X_+, \Sigma^m_{\mathbf{T}}(q\text{-}\mathrm{dR}^{\mathrm{mot}})^\wedge_p) \simeq R\Gamma_{\mathbb{\Delta}}(\mathfrak{X}_p/A) \otimes_A (\mathbf{L}^\wedge_p)^{\otimes -m}[2m].$$

In other words, $(q\text{-}\mathrm{dR}^{\mathrm{mot}})^\wedge_p$ represents relative prismatic cohomology over the q-de Rham prism.

Proof. Let Φ be the p-completion functor followed by the forgetful functor. As p-completion is the limit of the cofibers of p^k, Lemma 5.1(3) gives $\Phi\, q\text{-}\mathrm{dR}^{\mathrm{mot}} \simeq (q\text{-}\mathrm{dR}^{\mathrm{mot}})^\wedge_p$. By Lemma 5.1(2),

$$\mathrm{map}(X_+, \Sigma^m_{\mathbf{T}}(q\text{-}\mathrm{dR}^{\mathrm{mot}})^\wedge_p) \simeq \big(q\text{-}\mathrm{dR}(X) \otimes \mathbf{L}^{\otimes -m}[2m]\big)^\wedge_p \simeq q\text{-}\mathrm{dR}(X)^\wedge_p \otimes_A (\mathbf{L}^\wedge_p)^{\otimes -m}[2m],$$

where the second step uses that $\mathbf{L}$ is free of rank one. Now apply Theorem 5.5 to X and $\mathbf{P}^1$. □

Proposition 5.7. — *Let $S = \operatorname{Spec} \mathbf{Z}_p[\zeta_p]$, viewed as an A/I-scheme, let $f\colon S \longrightarrow \operatorname{Spec} \mathbf{Z}$ be the structure map, and let $\mathrm{H}A^{\mathbb{\Delta}} \in \mathrm{MS}_S$ be the motivic spectrum representing relative prismatic cohomology over the q-de Rham prism [AHI24b, Example 6.17]. Then there is a canonical isomorphism $\mathbf{L}^\wedge_p \cong A\{-1\}$, and an equivalence*

$$(q\text{-}\mathrm{dR}^{\mathrm{mot}})^\wedge_p \simeq f_*\mathrm{H}A^{\mathbb{\Delta}}$$

in $\mathrm{MS}_{\mathbf{Z}}$. In particular, the twists of $(q\text{-}\mathrm{dR}^{\mathrm{mot}})^\wedge_p$ are the Breuil–Kisin twists.

Proof. For $E \in \mathrm{MS}_S$ we have $(f_*E)_n(X) \simeq E_n(X_S)$, compatibly with bonding maps, since f^* is symmetric monoidal with $f^*X_+ \simeq (X_S)_+$. By [AHI24b, Construction 6.5, Example 6.17], the levels of $f_*\mathrm{H}A^{\mathbb{\Delta}}$ are therefore $X \mapsto R\Gamma_{\mathbb{\Delta}}(\mathfrak{X}_p/A)\{n\}[2n]$, and its bonding maps are multiplication by a class $c \in \widetilde{H}^2_{\mathbb{\Delta}}(\mathbf{P}^1_{A/I}/A)\{1\}$. Since these maps are equivalences, c generates this module, which is $\mathbf{L}^\wedge_p\{1\}$ by Proposition 5.6. This gives $\mathbf{L}^\wedge_p \cong A\{-1\}$, and hence natural levelwise equivalences $\phi_n\colon ((q\text{-}\mathrm{dR}^{\mathrm{mot}})^\wedge_p)_n \simeq (f_*\mathrm{H}A^{\mathbb{\Delta}})_n$.

Our bonding maps are also multiplication by a generator of the same rank one module (this uses that Theorem 5.5 is multiplicative). So the ϕ_n commute with the bonding maps up to units $u_n \in A^\times$. Replacing ϕ_n by $(u_0 \cdots u_{n-1})^{-1}\phi_n$ makes them commute on the nose, and Proposition 3.5 gives the claimed equivalence in $\mathrm{MS}_{\mathbf{Z}}$. □

We conclude by relating the different motivic spectra using a fracture square. For $M \in D(\mathbf{Z}[\![q-1]\!])^\wedge_{(q-1)}$, put

$$\Phi_{\mathbf{Q}}(M) := (M \otimes \mathbf{Q})^\wedge_{(q-1)}, \qquad \Phi_{\mathbf{A}}(M) := \Big(\prod_p M^\wedge_p \otimes \mathbf{Q}\Big)^\wedge_{(q-1)}.$$

This defines exact endofunctors of $D(\mathbf{Z}[\![q-1]\!])^\wedge_{(q-1)}$.[1] Composing with the forgetful functor to Sp, we write $q\text{-dR}^{\mathrm{mot}}_{\mathbf{Q}} := \Phi_{\mathbf{Q}}\, q\text{-dR}^{\mathrm{mot}}$ and $q\text{-dR}^{\mathrm{mot}}_{\mathbf{A}} := \Phi_{\mathbf{A}}\, q\text{-dR}^{\mathrm{mot}}$ for the motivic spectra of Lemma 5.1(2).

Theorem 5.8. — *There is a cartesian square*

$$\begin{array}{ccc} q\text{-dR}^{\mathrm{mot}} & \longrightarrow & \prod_p (q\text{-dR}^{\mathrm{mot}})^\wedge_p \\ \downarrow & & \downarrow \\ q\text{-dR}^{\mathrm{mot}}_{\mathbf{Q}} & \longrightarrow & q\text{-dR}^{\mathrm{mot}}_{\mathbf{A}} \end{array}$$

in $\mathrm{MS}_{\mathbf{Z}}$.

Proof. For every $M \in D(\mathbf{Z}[\![q-1]\!])^\wedge_{(q-1)}$, the arithmetic fracture square

$$\begin{array}{ccc} M & \longrightarrow & \prod_p M^\wedge_p \\ \downarrow & & \downarrow \\ M \otimes \mathbf{Q} & \longrightarrow & \prod_p M^\wedge_p \otimes \mathbf{Q} \end{array}$$

is cartesian in $D(\mathbf{Z}[\![q-1]\!])$, naturally in M. Derived $(q-1)$-completion is exact, and it does not change M or $\prod_p M^\wedge_p$. So the completed square underlies a cartesian square of exact functors $D(\mathbf{Z}[\![q-1]\!])^\wedge_{(q-1)} \longrightarrow D(\mathbf{Z}[\![q-1]\!])^\wedge_{(q-1)}$. By Lemma 5.1(3), it induces the cartesian square in the statement. □

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[1] Here the notation $\mathbf{A}$ refers to the ring of adeles, not the affine line or $\mathbf{A}^1$-invariance. Although this is unfortunate notation, it should not cause confusion.

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